

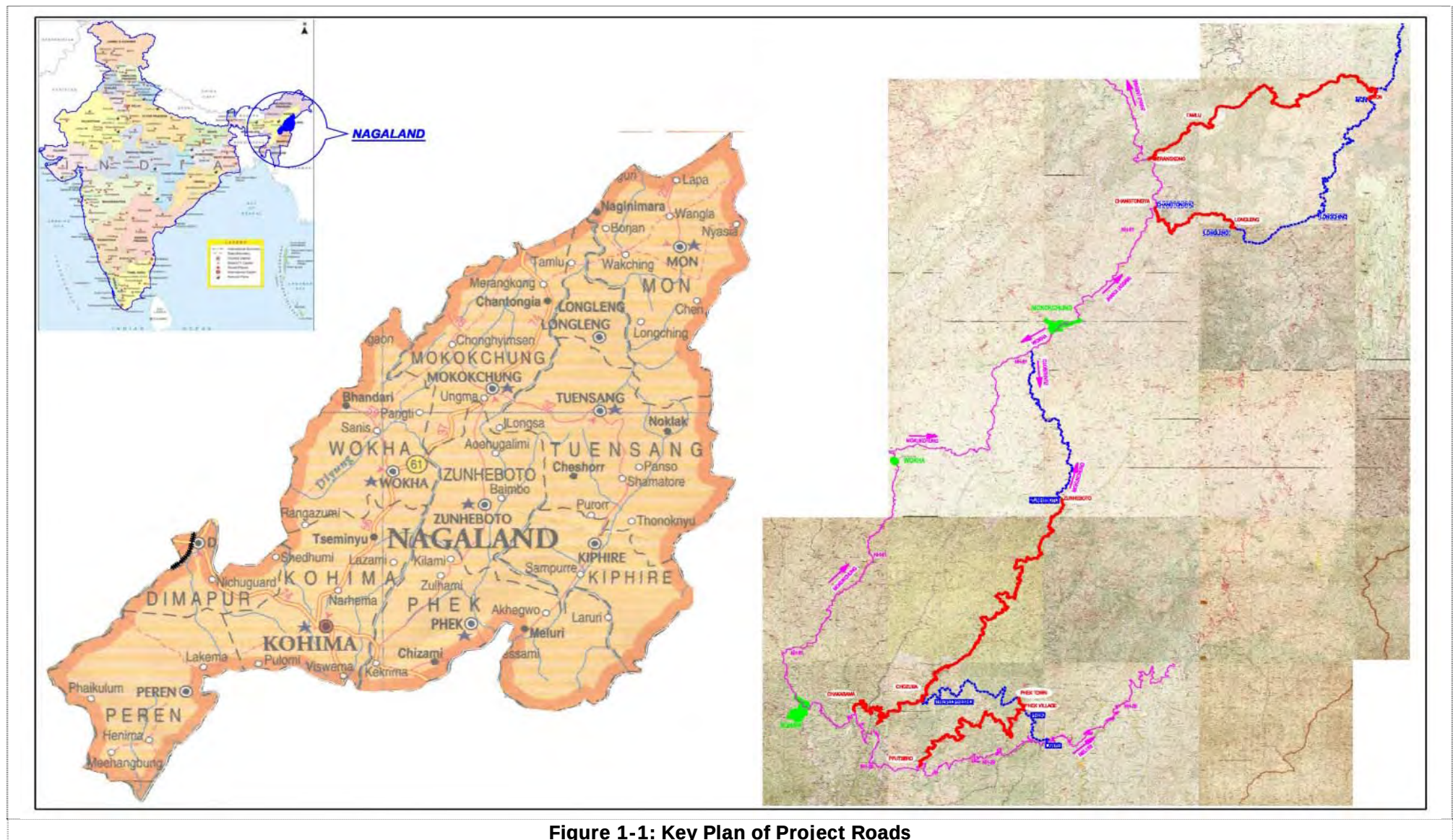


Figure 1-1: Key Plan of Project Roads

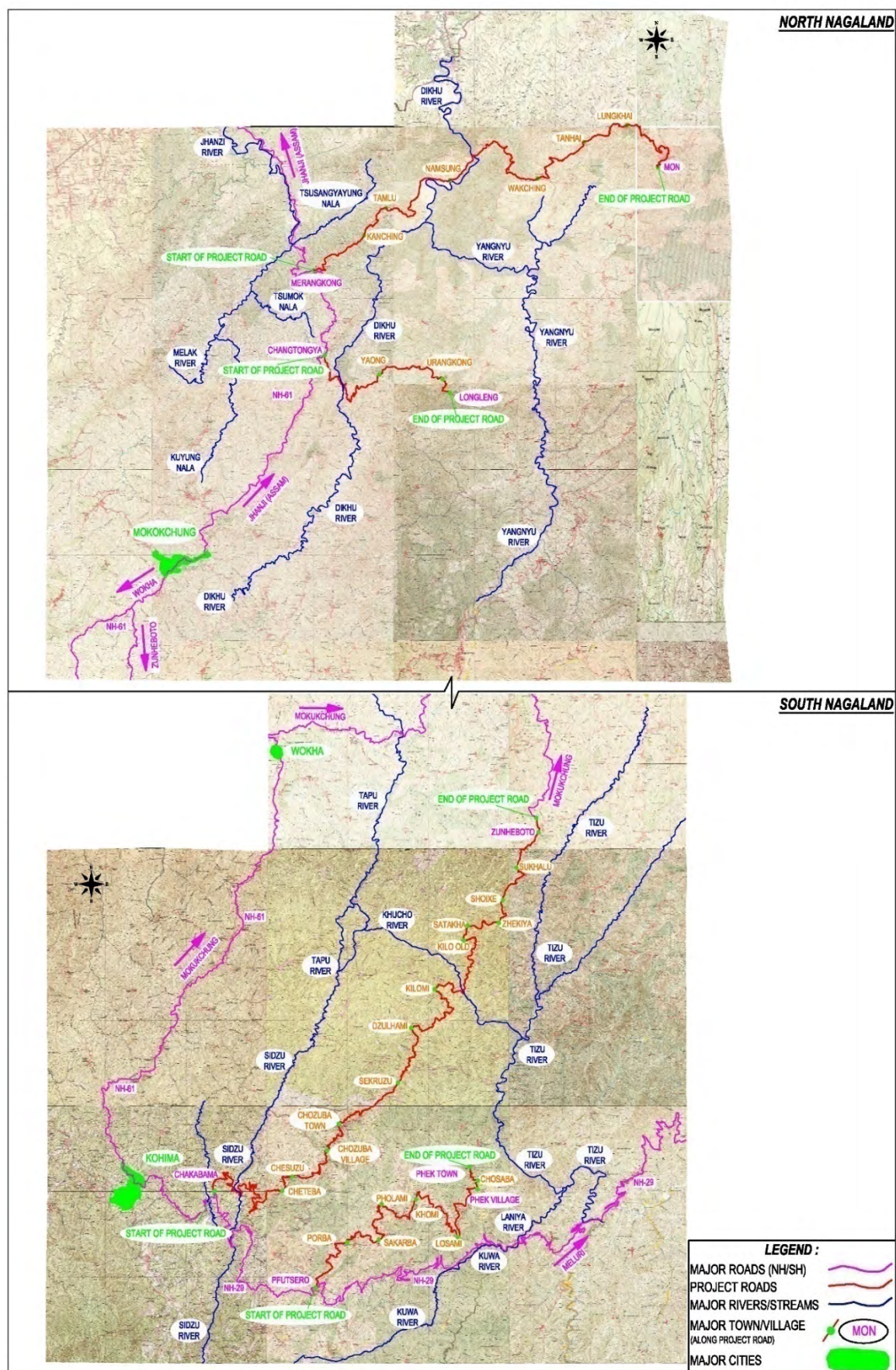


Figure 1-2: Project Roads

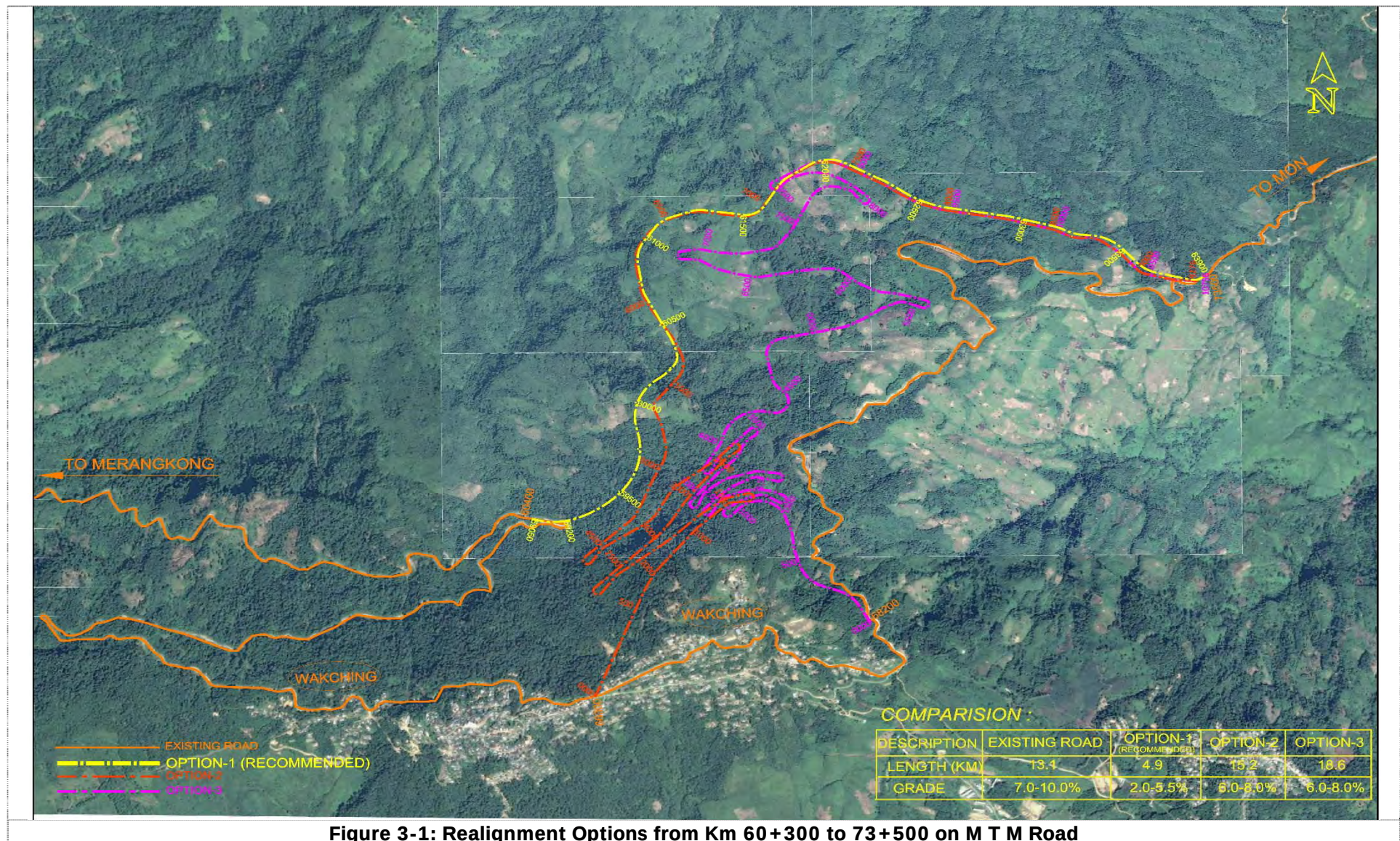


Figure 3-1: Realignment Options from Km 60+300 to 73+500 on M T M Road

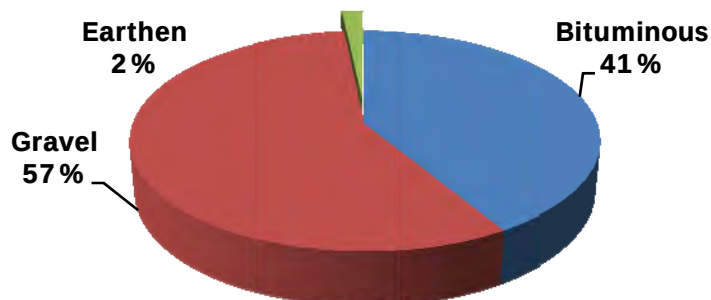
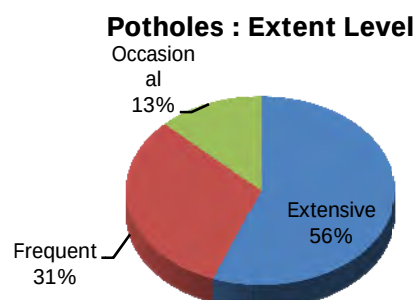
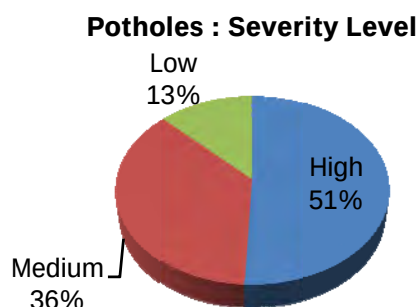
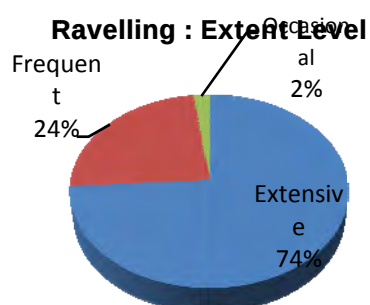
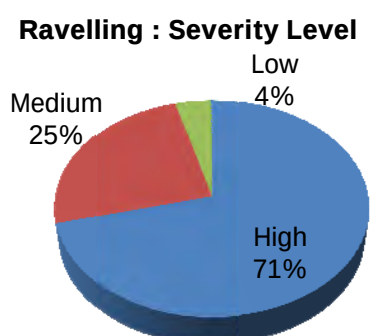
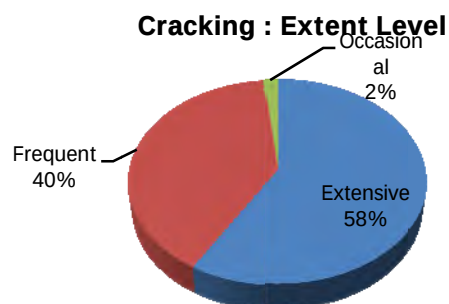
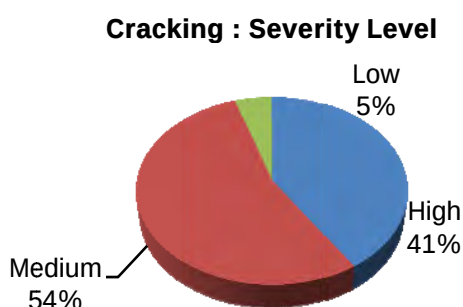


Figure 4-1: Surfacing Layer Distribution along M T M Road

Condition of Bituminous Surface

Bituminous surface distresses were measured in severity (Low/ Medium/ High) and extent (Occasional/ Frequent/ Extensive) level. The results of pavement condition survey for bituminous surfacing lengths are summarized and presented in pie charts format:



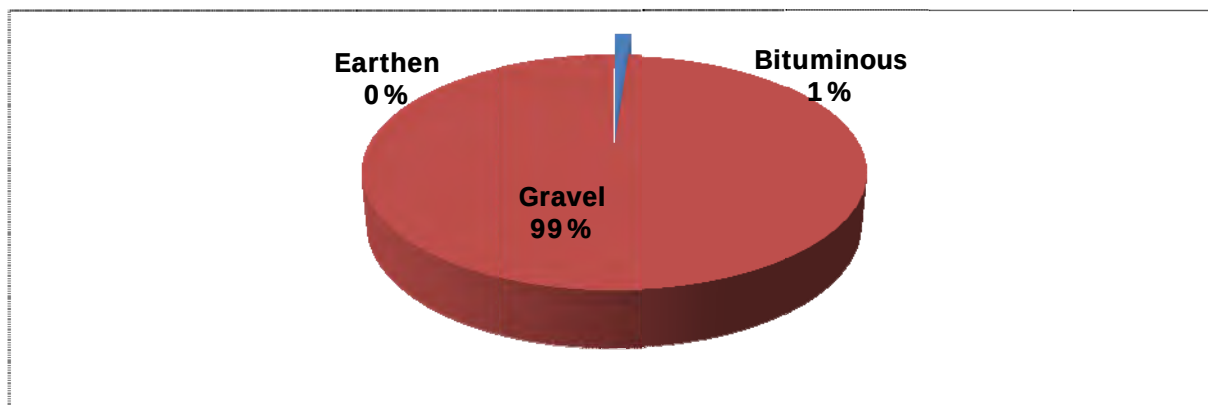


Figure 4-3: Surfacing Layer Distribution along C L Road

Shoulder

Existing Project Road Cross Section has earthen shoulder having 1.0 m width on both sides for 0.700km length which is in poor condition and for remaining 28.830 km length project road cross section does not possess any shoulder.

Drain

Unlined earthen drain exists on left or right side of road for 15.100 km length on Hill Side. Partially functional Open water channel exists on right side (Hill Side) of road for 0.530 km length in Longleng Town.

4.2.3. Chakabama – Zunheboto Road

Type of Carriageway Surface

The Project Road has Bituminous and Gravel Surfacing layer which is completely damaged. The existing pavement crust has seen his life and entire pavement crust including surfacing for entire road length would require new construction. In some part of project road, pavement is in under-construction.

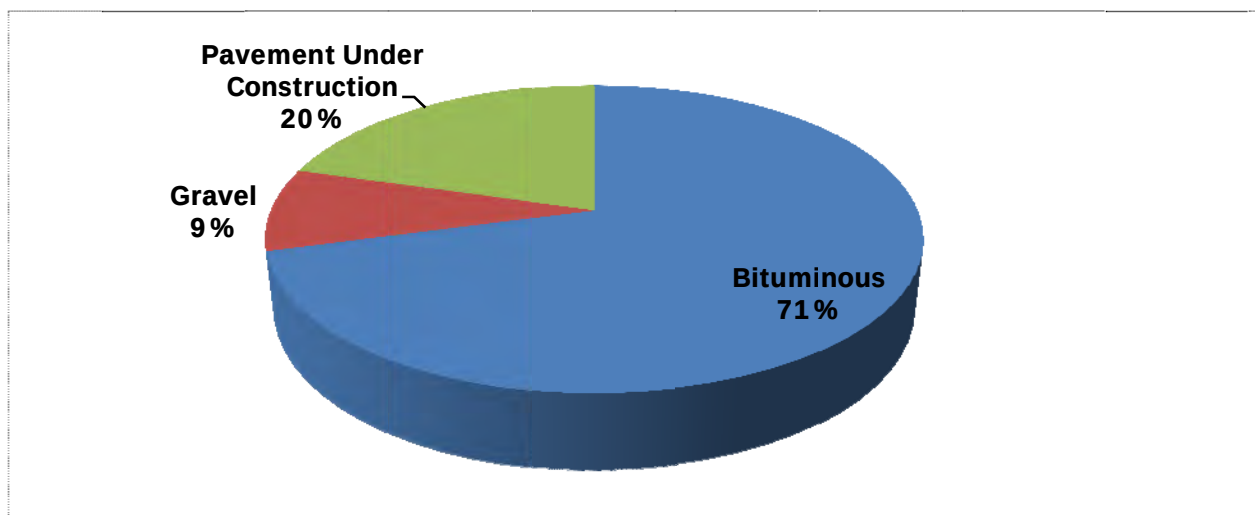
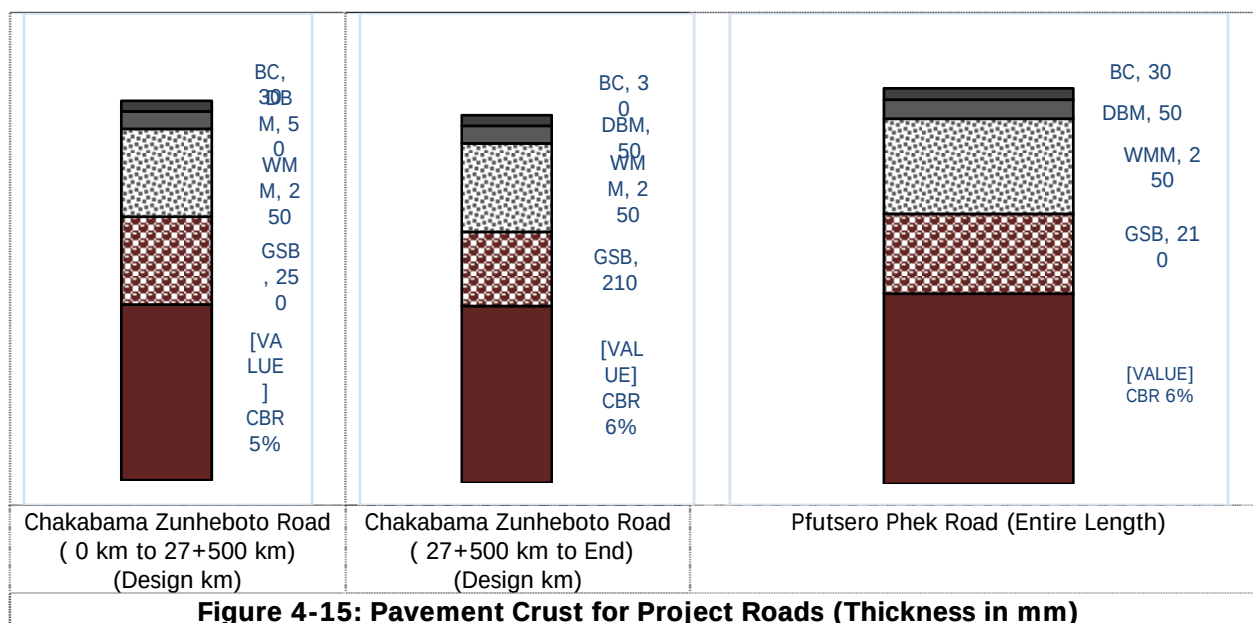
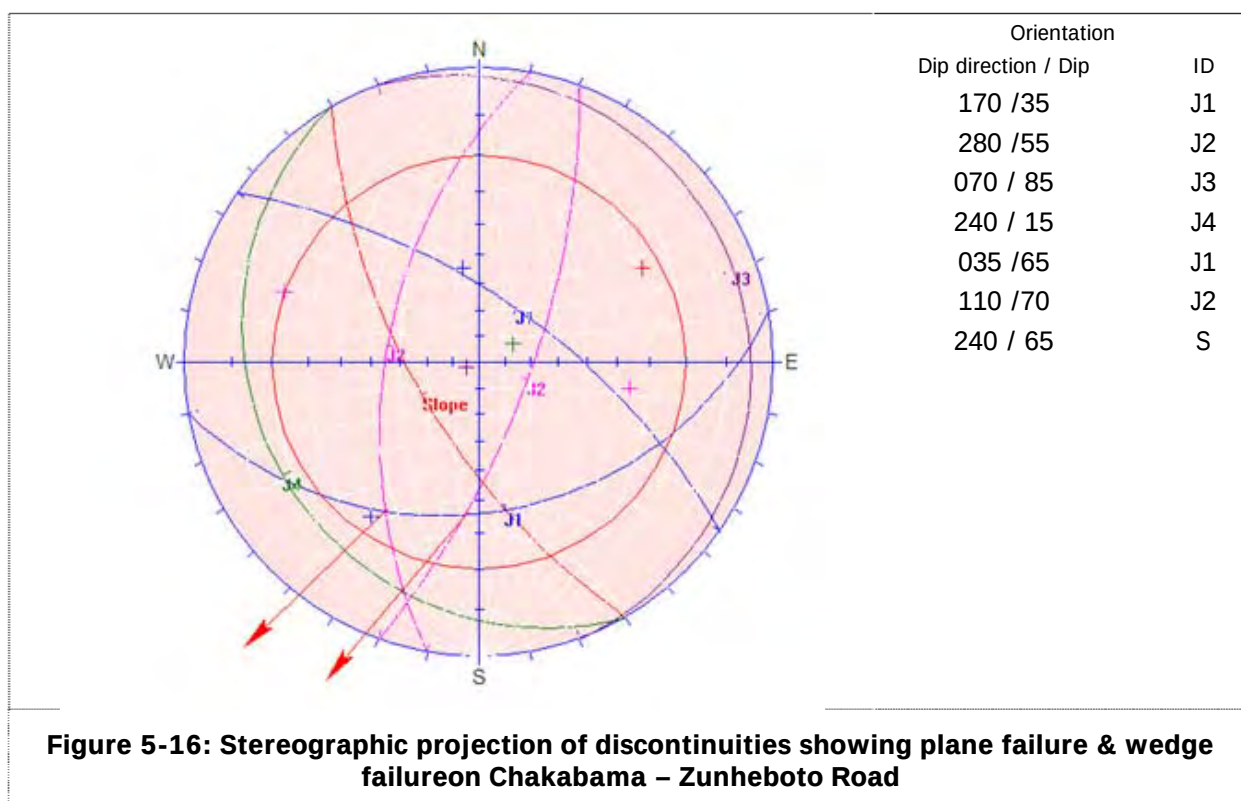
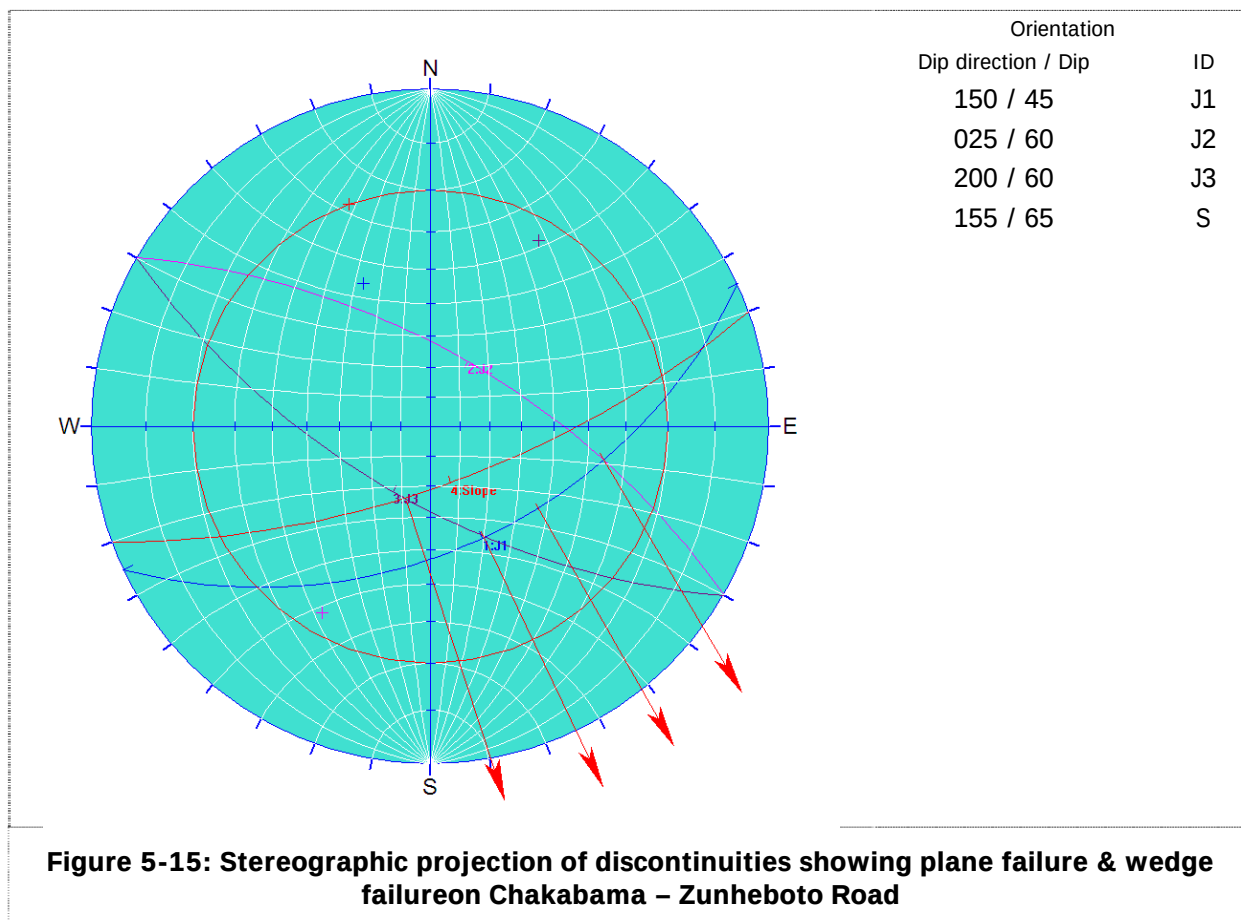


Figure 4-4: Surfacing Layer Distribution along C Z Road

Condition of Bituminous Surface

Bituminous surface distresses were measured in severity (Low/ Medium/ High) and extent (Occasional/ Frequent/ Extensive) level. The results of pavement condition survey for bituminous surfacing lengths are summarized and presented in pie charts format:





TCS I (b): TCS for Project Road Sections requiring Fill on Valley Side

This Typical Cross Section includes following cross sectional elements

- Carriageway Width as 7.0 m;
- Paved Shoulder on Both Sides = 1.5 m;
- Trapezoidal Random Rubble Masonry Drain on hill Side with base width of 0.3 m and depth varying from 0.5 m having thickness of masonry as 0.2 m.
- Slope Stability / Erosion Control Preventive Measures on Hill Side along with a Catch Water Drain in Top Most Cut Slope;
- Retaining / Reinforced Earth Wall on Valley Side with Parapet Wall;
- Total Roadway Width = 12.5 m

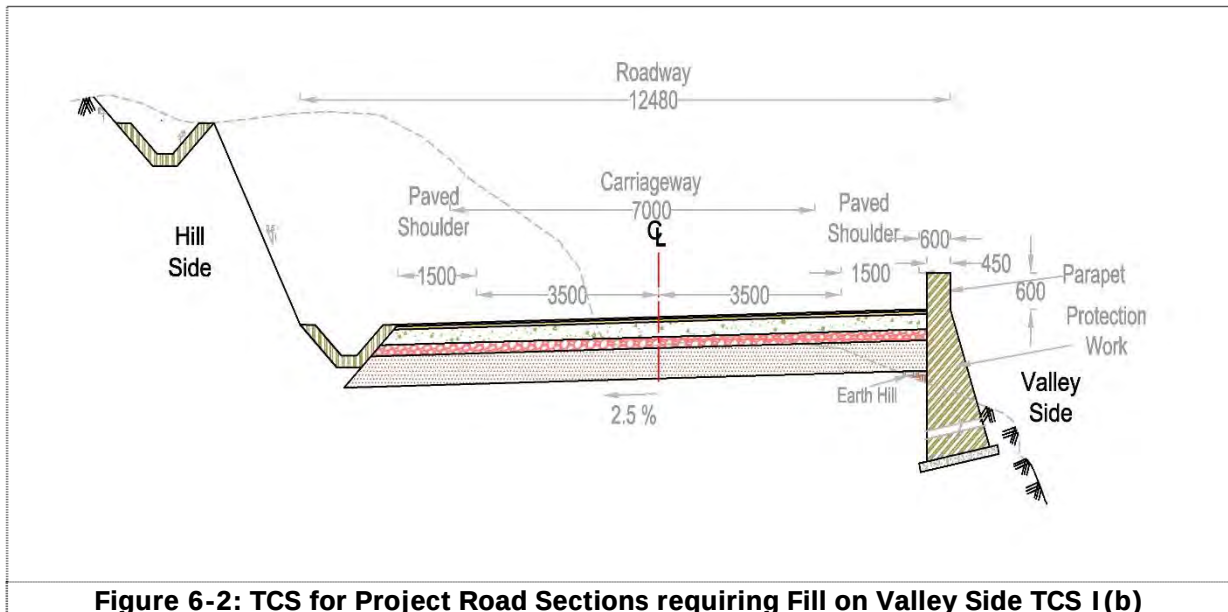


Figure 6-2: TCS for Project Road Sections requiring Fill on Valley Side TCS I(b)

TCS II: Typical Cross Section for project road section on ridge

This Typical Cross Section includes following cross sectional elements

- Carriageway Width as 7.0 m;
- Paved Shoulder on either Side = 1.5 m;
- Trapezoidal Random Rubble Masonry Drain on hill Side with base width varying from 0.3 m to 0.5 m and depth varying from 0.5 m to 0.6 m having thickness of masonry as 0.2 m.
- In general Height of Finished Road Level from ground is 1 m; and
- Total Roadway Width = 19 – 24 m.

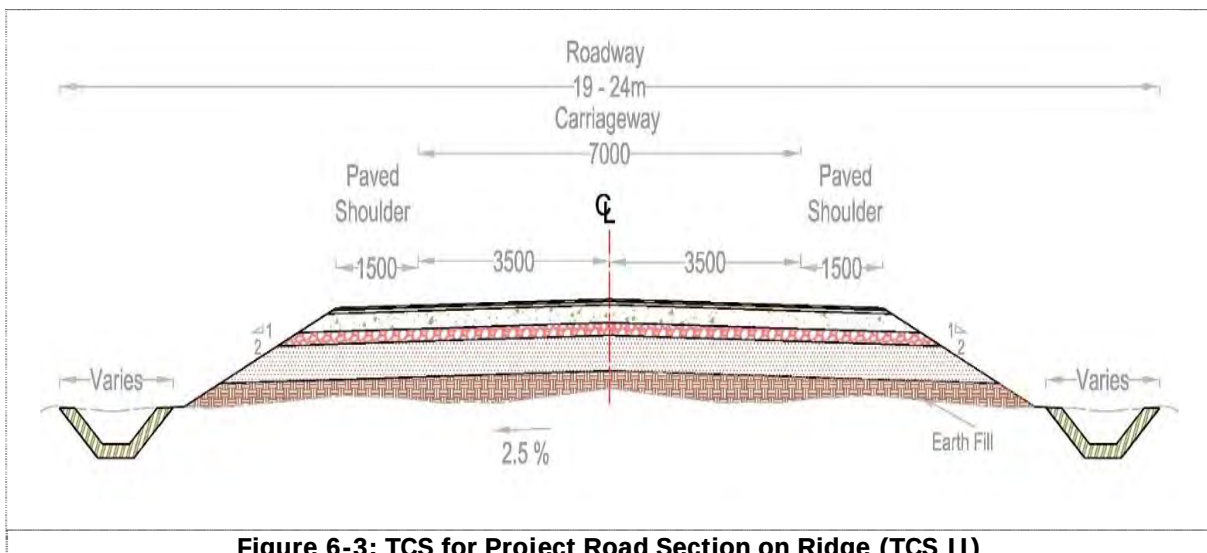


Figure 6-3: TCS for Project Road Section on Ridge (TCS II)